



Mayank Jha
Computer Science & Engineering
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M.Tech.
Gender: Male
DOB: 23/09/2002

Examination	University	Institute	Year	CPI / % Credits
Post Graduation	IIT Bombay	IIT Bombay	2027	null
Graduation	IIIT Delhi	IIIT Delhi	2024	7.11
Graduation Specialization: Computer Science and Biosciences				

M.TECH PROJECT AND SEMINAR

Game Theoretic Approaches for Security Planning

July 2026 - Present

Prof. Swaprava Nath | Dept. of CSE | M.Tech Project

IIT Bombay

- Studied **game-theoretic decision-making in defence** through a 28-paper review and 3 frameworks, exploring minimax, Counterfactual Regret Minimization, and Online Learning to understand how defenders adapt against an adaptive adversary in SAM-SEAD engagements
- Modeled a **22-aircraft, 10-barrier** patrol problem as a resource-allocation optimization game, deriving optimal coverage through a 3-stage LP and mixed-strategy formulation under a **1,210-hour/month** flying constraint and exposing a **2× capacity gap** for guaranteed full coverage
- Converted the **optimal allocation into a concrete patrol schedule** using LP duality and column generation to construct an integer mixed strategy, randomized phase offsets to generate unpredictable patrol timetables; proved $\leq K + 1$ iteration termination and worst-case optimality

PROJECTS

Real-Time Order Matching Engine

Jan 2026 – May 2026

Prof. G Sivakumar, Dept. of CSE, Course Project

IIT Bombay

- Implemented deterministic **price-time-priority matching** for MARKET/LIMIT orders and partial fills using heap-based order books; benchmarked the matching core at **4.72M orders/sec** with **42 ns p50** and **500 ns p99** latency against a 500K-order resting book
- Architected an **end-to-end trading platform** with a Rust/Tokio matching service, Express REST→gRPC gateway, React trading dashboard and Keycloak OIDC authentication, unified through shared Protobuf contracts across Rust, Node.js and test tooling
- Decoupled matching from ClickHouse I/O through an asynchronous mpsc storage worker, enabling crash recovery via snapshot replay and **100% traffic mirroring** to a shadow engine through nginx; instrumented the pipeline with OpenTelemetry tracing and metrics

Search & Retrieval Engine

July 2026 – Present

Prof. Soumen Chakrabarti, Dept. of CSE, Course Project

IIT Bombay

- Engineered an end-to-end search engine spanning a concurrent Python crawler, C++ indexing engine, and FastAPI query service; implemented custom tokenization and a **flat-array positional inverted index** with LEB128-encoded positions and from-scratch **multi-term BM25** scoring across 20K documents, 8M+ postings, and 850K+ unique terms
- Built a hybrid semantic retrieval pipeline integrating 384-dimensional sentence-transformer embeddings with BM25 and PageRank through min-max normalized weighted fusion, improving real-corpus **nDCG@10 from 0.46 to 0.63** while automatically identifying and quantifying a **~58× PageRank selection bias** in relevance judgments
- Designed a byte-exact cross-language binary index format with **LEB128 delta encoding** and versioned sections, preserving compatibility across an $81\times$ corpus expansion with zero downstream code changes; containerized the pipeline and calibrated 2–5GB memory limits, establishing a stable 5GB bound with $\sim 730\text{MB}$ headroom under real query load

PostgreSQL Index Advisor

July 2026 – Present

Prof. S. Sudarshan, Dept. of CSE, Course Project

IIT Bombay

- Engineered a Java 21/Spring Boot index advisor that screens candidates with **HypoPG** and validates winners using real **CREATE INDEX + EXPLAIN ANALYZE**; live production testing measured up to **96.5%** query improvement
- Diagnosed a **HikariCP concurrency race** and cache-warming measurement bias through concurrent-load testing and symmetric median-of-3 benchmarking; secured dynamic DDL with schema allowlisting and quoted identifiers
- Implemented greedy **marginal-utility index-set selection** with B-tree prefix-aware domination pruning, eliminating redundant indexes across workloads; HypoPG screen-then-validate benchmarking delivered **~1.2×** average speedup across 5 workloads

NETRA: Full-Stack News Aggregation & Personalization Platform

July 2025 – Nov 2025

Prof. Om Damani, Dept. of CSE, Course Project

IIT Bombay

- Engineered a **full-stack news platform** with Flask, React and PostgreSQL/pgvector, integrating JWT authentication, relational integrity and **127 backend tests at 89% coverage**; deployed end-to-end across Supabase, Render and Vercel
- Built a **384-dimensional semantic recommendation pipeline** using sentence-transformer embeddings with cosine similarity, recency decay, category affinity and engagement signals; evaluated ranking quality with **nDCG@5 of 0.241** on a 2,027-article corpus
- Optimized data retrieval through **bulk SQL aggregation**, reducing per-page database queries from 80 to 2, while implementing an $O(nk)$ greedy story-clustering algorithm with **0.35s runtime** and debugging multi-worker scheduler failures

Distributed Key-Value Cache

Jan 2024 – July 2024

Self Project

- **Engineered a distributed in-memory key-value cache from scratch**, implementing a TCP protocol, fixed-thread-pool server, LRU eviction, and TTL expiration; a 3-node cluster sustained **6,000/6,000 successful operations at 20 concurrent clients**, versus 2,149/6,000 (36%) on 1 node
- **Diagnosed the load collapse to TCP ephemeral-port exhaustion**, using `netstat` to identify 16,340 `TIME_WAIT` sockets; benchmarked 6 load scenarios, measuring **1.7× higher throughput** with 3-node partitioning at 10 clients
- **Hardened correctness by closing 3 race conditions** in capacity checks, TTL updates, and LRU state, using `ReentrantLock` to protect `LinkedHashMap`'s mutating-on-read access order; validated with **26 unit, protocol, and integration tests**

Distributed Genomic Streaming Pipeline

Jan 2024 – July 2024

Self Project

- **Found and fixed a hidden flaw** in a similarity-matching system built from scratch using Bloom filters and MinHash for scanning DNA sequencing data; discovered the similarity check breaks down when comparing a short DNA fragment against an entire genome and replaced it with an exact-match technique, **cutting Bloom candidates by 96% (6,274 → 244)** across 1.35M real sequencing reads
- **Trained a 19-feature PyTorch neural network** to classify flagged DNA sequences, testing against realistic sequencing errors and a wide range of simulated infection levels; achieved a **97.6–100% detection rate with 0 false positives**
- **Built a fault-tolerant streaming pipeline with Apache Kafka** to ingest DNA sequencing data across 8 parallel partitions with crash-safe PostgreSQL persistence; proved **12s recovery with zero message loss** across 20K reads under container-kill recovery tests in Docker

Multi-Model Content Moderation & NLP Platform

July 2023 – Nov 2023

Self Project

- Engineered an NLP classification pipeline over **7.7M chat messages**, evaluating 8 classical models, a recurrent model, and **DistilBERT** on an identical 200K balanced split; achieved a **22.5% relative F1 gain** over the best classical baseline
- Developed a **13-class fallacy analysis engine** with structure-aware preprocessing, LIME, multi-seed evaluation, and **1,000-resample bootstrap testing**; achieved a **6× gain in paraphrase robustness**, reducing prediction flips from **13.7% to 2.3%**
- Architected a **multi-model FastAPI serving layer** with per-model failure isolation, graceful fallback, model discovery, and health endpoints across two task-specific APIs; containerized with **Docker** and validated via end-to-end HTTP execution

TECHNICAL SKILLS

Languages, Tools and Frameworks: Java, C++, Python, SQL, JavaScript, React, TypeScript, Redux, Vite, Tailwind

Backend & APIs: Spring Boot, Node.js, FastAPI, Flask, REST APIs, API Design

Databases: PostgreSQL, MySQL, Redis, Query Optimization, Indexing, pgvector

Distributed Systems: Kafka, Partitioning, Consumer Groups, Fault Tolerance, Backpressure, Idempotency, Offset Management

Systems & Engineering: Multithreading, Concurrency, TCP Sockets, Locks, Client-Server Architecture, LRU Cache, TTL Caching, Docker, Linux, Git, CI/CD, Debugging, Performance Optimization

Algorithms & Data Structures: Data Structures, Algorithms, Graph Algorithms, Dynamic Programming, Greedy Algorithms, String Algorithms, Searching, Optimization

COMPETITIVE PROGRAMMING

CodeChef Rating: 1450+ | Codeforces Rating: 1100+ | LeetCode Contest Rating: 1500+ & 270+ Problems Solved

POSITIONS OF RESPONSIBILITY

Teaching Assistant, CS101: Computer Programming and Utilization, IIT Bombay Aug 2025 – Nov 2025

Personally Guided 6 students through C++ fundamentals via doubt-clearing sessions resulting in improved course performance.

Teaching Assistant, CS218: Design and Analysis of Algorithms, IIT Bombay Jan 2026 – May 2026

Prepared 4 quizzes, designed rubrics, supervised examinations, and graded 100+ evaluation papers to support course assessment.

Teaching Assistant, CS6001: Game Theory & Algorithmic Mechanism Design, IIT Bombay July 2026 – Present

RELEVANT COURSEWORK

IIT Bombay: Software Engineering, Indexing and Retrieving Text and Graphs*(ongoing course), Foundations of Machine Learning

IIIT Delhi: Operating Systems, Data Structures and Algorithms, Fundamentals of Database Management Systems, Computer Organisation, Digital Circuits

EXTRACURRICULARS

Community Work with E-Vidyaloka | Sports & Fitness: Recreational Sports and Regular Strength Training

HOBBIES AND INTERESTS

Playing Chess, Reading, Learning Philosophy & Psychology, Exploring Scriptures, Spirituality & Self-Reflection, Exploring Game Theory, International Relations, Economics & History, Building Tools to Solve Everyday Problems, Cinephile, Music Enthusiast